F12 4PA CAR

Lecture 11: Section 1.8
Combinations of Functions

Algebra of functions

Composition of functions

Decomposing a composite function

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Algebra of Functions

Let f and g be two functions with domains A and B. We define some new functions:

Name	Definition	Domain
$f \pm g$	$f(x) \pm g(x)$	$A \cap B$
mult. fg	$(fg)(x) = f(x) \cdot g(x)$	$A \cap B$
$\frac{f}{g}$	$\frac{f}{g}(x) = \frac{f(x)}{g(x)}$	$A \cap B \cap \{x \mid g(x) \neq 0\}$

ex. Given $f(x) = x^3$ and $g(x) = \sqrt{x}$. Find the function $\frac{f}{g}$ and its domain.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^3}{\sqrt{x}} = x^{3-\frac{1}{2}} = x^{5/2}$$

$\boxed{D: x > 0}$ or $(0, \infty)$

D: f is all reals

D: g is $x \geq 0$, note $x \neq 0$, since 0 in denom.

$\sqrt{9} + \sqrt{4} \neq$ [redacted] $= \sqrt{9+4} = \sqrt{13}$

$3 + 2 = 5$

$1 - x \geq 0$

$1 \geq x$

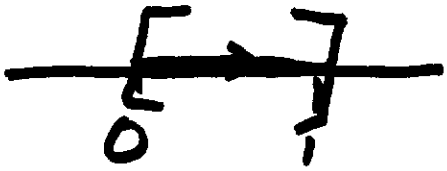
ex. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, find the function $f+g$ and its domain.

$(f+g)(x) = f(x) + g(x) = \boxed{\sqrt{x} + \sqrt{1-x}}$

D: f is $x \geq 0$

D: g is $x \leq 1$

$\boxed{[0,1]}$



ex. If $f(x) = \dfrac{1}{x+1}$ and $g(x) = \dfrac{x}{x+1}$, find each function and its domain.

D: $x \neq -1$

D: $x \neq -1$

1) $fg = (f \cdot g)(x) = f(x) \cdot g(x) =$

$= \dfrac{1}{x+1} \cdot \dfrac{x}{x+1} = \boxed{\dfrac{x}{(x+1)^2}}$

D: $x \neq -1$

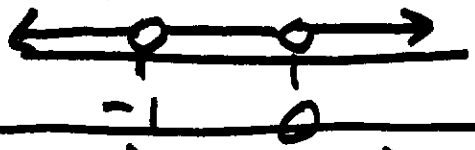
$\boxed{(-\infty, -1) \cup (-1, \infty)}$

2) $\dfrac{f}{g} = \left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)} =$ [redacted] $= \dfrac{\frac{1}{x+1}}{\frac{x}{x+1}} =$

$= \dfrac{1}{\cancel{x+1}} \cdot \dfrac{\cancel{x+1}}{x} = \boxed{\dfrac{1}{x}}$

D: f is $x \neq -1$

D: g is $x \neq -1$



$\boxed{(-\infty, -1) \cup (-1, 0) \cup (0, \infty)}$

also $g(x) \neq 0$, thus $x \neq 0$

Composition of Functions

Def. Given two functions f and g, the **composition** of function f [redacted] with function g is defined by

f composed with g

$$(f \circ g)(x) = f(g(x))$$

The domain of $(f \circ g)$ is the set of x in the domain of g such that $g(x)$ is in the domain of f.

$$x \longrightarrow \boxed{g} \xrightarrow{g(x)} \boxed{f} \longrightarrow f(g(x))$$

inside function

$x - 3 \geq 0$

$x \geq 3$

ex. Let $f = x^2$ and $g = \sqrt{x-3}$. Find the functions $f \circ g$ and $g \circ f$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{x-3}) = (\sqrt{x-3})^2 = \boxed{x-3}$$

D of g is $x \geq 3$

D is all reals

thus $\boxed{[3, \infty)}$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = \boxed{\sqrt{x^2-3}}$$

D: all reals

D

D: $x^2 - 3 \geq 0$

$(x+\sqrt{3})(x-\sqrt{3}) \geq 0$

$-\sqrt{3}$ $\sqrt{3}$

test $x=-10$ | $x=0$ | $x=10$

$-\cdot-=$ | $+\cdot-=$ | $+\cdot+=$

$\oplus$ | $\ominus$ | $\oplus$

$-\sqrt{3}$ $\sqrt{3}$

$$\boxed{(-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)}$$

*Domain of composite function.
Intersection of argument $x \neq 1$
and final solution

ex. If $f(x) = \dfrac{1}{x+2}$ and $g(x) = \dfrac{4}{x-1}$, find the following functions and their domains.

1) $f \circ g = f(g(x)) = f\left(\dfrac{4}{x-1}\right) =$

$$\frac{1}{\frac{4}{x-1}+2} = \left(\frac{1}{\frac{4}{x-1}+2}\right)\frac{(x-1)}{(x-1)} = \frac{x-1}{4+2(x-1)} = \boxed{\frac{x-1}{2x+2}}$$

$2x+2 \neq 0$
$2x \neq -2$
$x \neq -1$

-1 $\quad$ 1

$\boxed{(-\infty,-1)\cup(-1,1)\cup(1,\infty)}$

2) $f \circ f$

$$(f \circ f)(x) = f(f(x)) = f\left(\frac{1}{x+2}\right) =$$

$$\frac{1}{\frac{1}{x+2}+2} = \frac{1}{\frac{1}{x+2}+2\left(\frac{x+2}{x+2}\right)} = \frac{1}{\frac{1+2(x+2)}{x+2}} = \frac{1}{\frac{2x+5}{x+2}} = \boxed{\frac{x+2}{2x+5}}$$

$2x+5 \neq 0$
$2x \neq -5$
$x \neq \frac{-5}{2}$

NOTE: The domain of $F(x) = (f \circ g)(x)$ is the intersection of the domain of inner function g and the resulting function F.

$-\frac{5}{2}$ $\quad$ -2

D: $x \neq -2$ also $x = -\frac{5}{2}$

$\boxed{(-\infty, -\frac{5}{2})\cup(-\frac{5}{2},-2)\cup(-2,\infty)}$

Optional Material.

A Composition of Three Functions

If f, g and h are functions, then

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

ex. If $f(x) = \dfrac{1}{x}$, $g(x) = x^2 - 5$, and $h(x) = \sqrt{x}$,

find $f \circ g \circ h$ and its domain.

$f(g(h(x))) =$

$f(g(\sqrt{x})) =$

$x \geq 0$

$x = 5$

$f(\sqrt{x}^2 - 5) = f(x-5) = \boxed{\dfrac{1}{x-5}}$

D: $\boxed{[0, 5) \cup (5, \infty)}$

Decomposing a Composite Function

To decompose a composite function $(f \circ g)(x)$ is to identify two functions f and g that make up a given composite function, where $g(x)$ is the inner function and $f(x)$ is the outer function. Many correct answers!

inner function

<u>ex.</u> Given $F(x) = (3x-1)^8$, find functions f and g such that $F = f \circ g$. $= f(g(x)) = f(3x-1) = (3x-1)^8 = F$

$$\boxed{\begin{aligned} g(x) &= 3x-1 \\ f(x) &= x^8 \end{aligned}}$$

or $g(x) = (3x-1)^2$

$f(x) = x^4$

$f(g(x)) = f((3x-1)^2) = ((3x-1)^2)^4$

$= (3x-1)^8 = F$

outer function

<u>ex.</u> Find two functions f and g such that $(f \circ g)(x) = \dfrac{1}{\sqrt{2x^2-1}}$.

inner funct.

$g(x) = \sqrt{2x^2-1}$

$f(g(x)) = f(\sqrt{2x^2-1})$

$f(x) = \dfrac{1}{x}$

$= \boxed{\dfrac{1}{\sqrt{2x^2-1}}} = (f \circ g)(x)$

outer funct.

NOTE: There are many correct answers!

Applications

ex. The weekly cost C of producing x units is given by $C(x) = 60x + 750$. The number of units x produced in t hours is given by $x(t) = 50t$.

1) Find and interpret $(C \circ x)(t) =$

$C(x(t)) = C(50t) = 60(50t) + 750 = 3000t + 750$

cost in terms of t hours

2) Find the time that must elapse in order for the cost to increase to \$15,000. let $C = 15{,}000$

$C = 3000t + 750$

$15{,}000 = 3000t + 750$

$\vdots$

$\boxed{t = 4.75 \text{ hours}}$

ex. An oil tanker is leaking, causing a circular pool of oil with an expanding radius r given by $r(t) = \sqrt{t+1}$, where t is the number of minutes after the leak was first discovered and r is measured in meters.

$A = \pi r^2$

1) Find a formula that gives the area A of the circular region after t minutes.

we know

$A(r) = \pi r^2$ also $r(t) = \sqrt{t+1}$

$(A \circ r)(t) = A(r(t)) = A(\sqrt{t+1}) = \pi(\sqrt{t+1})^2$

$\boxed{A = \pi(t+1) \text{ sq. meters}}$

$A(t) = \pi(t+1)$

2) Find the area of the spill when it was first discovered. Let $t=0$

$A(0) = \pi(0+1) = \pi$ sq. meters

3) What is the area three minutes later?

$A(3) = \pi(3+1) = \boxed{4\pi \text{ sq. meters}}$

Practice.

Given $f(x) = \dfrac{x}{x-1}$ and $g(x) = -\dfrac{4}{x}$. Find each function and its domain.

1) $f \circ g$

2) $g \circ f$

3) $f \circ f$

4) $g \circ g$

Answer. 1) $f \circ g = \dfrac{4}{4+x}$, $x \neq -4, 0$ 2) $g \circ f = \dfrac{-4(x-1)}{x}$, $x \neq 0, 1$

3) $f \circ f = x$, $x \neq 1$ 4) $g \circ g = x$, $x \neq 0$